TEAM ID – 262045

BAJA SAEINDIA CAE EVALUATION

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EIMMORTALS

ABSTRACT

This report is a summary of the analysis team EIMMORTALS has done on the Baja SAEINDIA 2026 ATV. It consists of the structural analysis of the All-Terrain Vehicle including the roll cage and all the other system components associated with it. The 3-D solid model of the roll cage and components were generated using SolidWorks 2025 and the analysis was conducted on ANSYS Workbench 2025 R2. Our main objective of the analysis was to design an ATV that is highly competent without compromising on the safety aspects.

INTRODUCTION

The primary objective of our analysis was to not only create a structure that protects the driver in case of a collision but also supports the other subsystems like suspension, steering, braking, and etc. This report focuses on evaluating the ATV's key structural elements to ensure they meet the required safety margins, driver Ergonomics and improvements were made to meet maximum factor of safety possible (FOS).

Design Considerations

The design considerations that were considered were material selection, safety, frame design and CAE methodology.

Material Selection & Properties

We will use AISI 4130 Chromoly tubes for the roll cage and control arms with sizes specified in the rulebook that are primary members of OD 29.2 mm and secondary members of OD 25.4 mm. AISI 4130 is the best choice in our opinion because of its high strength-to-weight ratio compared to other available choices, ease of machinability and availability. Which makes it a perfect choice for Baja SAEINDIA competition.

Other parts such as uprights, wheel hub rotor, and more are of mild steel because of standardization of these parts at local OEMs these would definitely add some weight to the vehicle but they will be easy to replace or repair on breakdowns if it happens.

MATERIAL	AISI 1018	AISI 1020	AISI 4130
YIELD STRENGTH	370 MPa	294 MPa	460 MPa
TENSILE STRENGTH	440 MPa	394 MPa	560 MPa
ELONGATION	15%	36.5%	21.50%
COMPOSITION	C, Fe, Mn, P, S	C, Fe, Mn, P, S	C, Fe, Mn, P, S, Si, Mo

CAE METHODOLOGY

CAE analysis of a part goes through a series of processes before we get the analysis and it can be broken down into several stages.

1. **Geometry Cleanup/Assumptions**: after importing a geometry from CAD software like SolidWorks, it is important to refine and fix imperfections such as missing faces, gaps, overlapping surfaces, and unnecessary features that can hinder mesh generation and compromise simulation results. It is a critical step for preparing complex models for accurate and efficient finite element analysis.

2. **Subsystems**: For this report purposes we will be performing analyses on the following:

Component	Analyses
Roll Cage	Modal, Static
Uprights	Static, Cornering, Bump, Steering
Control Arms	Static
Brake Discs	Thermal
Wheel Hub Rotor	Torsion
Brake Pedal	Static

3. **<u>Mesh Specification</u>**: Keeping accuracy and precision as our focus we used the following mesh specifications

Subsystem	Element Size (mm)	Element Order	Element Type
Roll Cage	3–5 mm	Linear	Tetrahedral (where applicable)
Other Subsystems	3–5 mm	Quadratic	Tetrahedral

<u>Element Quality Checks</u>

To Ensure that we did achieve high quality of meshing we incorporated data from the graphs of mesh skewness and Jacobian.

Verification	Target Value	% of Elements Meeting Criteria
Skewness	< **0.5**	Most regions maintained below 0.5
Jacobian Ratio	≈ **1 (ideal)**	Majority of elements close to 1

Only after confirming that these quality checks were satisfied did we proceed with the final simulations.

Our roll cage was imported as vertex geometry in Ansys hence we cannot get a skewness or Jacobian graph with linear meshing but it is still good enough for our initial analysis. Other subsystems have high quality mesh by meeting each criterion.

<u>FEA</u>

<u>Static Analysis of Roll Cage</u>

1. <u>Front impact</u>

We did this analysis to test our Roll cage on front direct impact.

Input parameters:

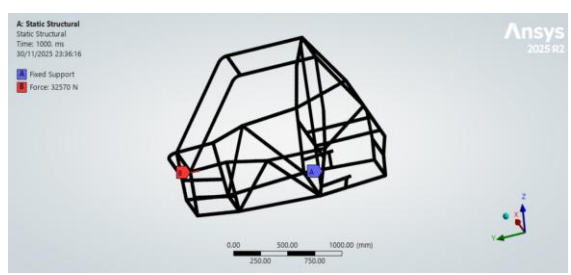
- Total weight of the vehicle with driver was considered to be 320 kgs
- The maximum speed of 55Km/h or 15.27m/s.
- The impact time for front impact was considered as 0.15 seconds

Boundary Conditions:

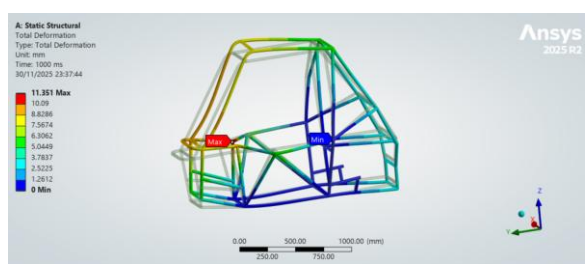
- The suspension points are fixed
- The force is applied on the forward most members of the chassis i.e. nose member

Calculations:

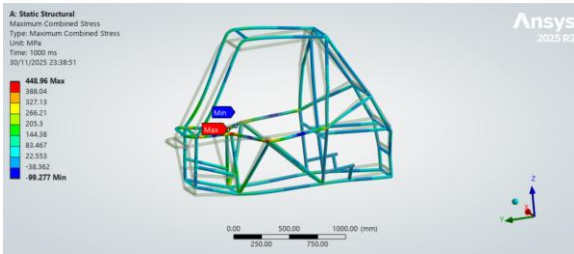
- Mass of the ATV= 320 kg
- Impact time= 0.15 sec
- speed= 55km/hr. or 15.27 m/s
- We know that Force = mass x acceleration
- Therefore, F= 320 x (15.27/0.15) =32570 N



Boundary Conditions



Total Deformation



Total Combined Stress

The Maximum Combines stress induced is 448.96 MPa. maximum deformation of 11.35 mm.

2. Rear impact

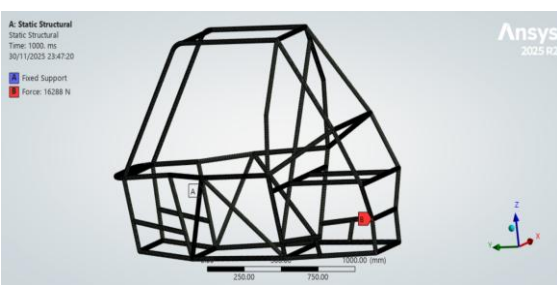
We did this analysis to simulate a scenario where the vehicle is hit by another vehicle from behind. Under such conditions, the forces act on the rear most members of the chassis.

Input Parameters:

- Weight 320kgs
- Speed 55km/hr. or 15.27 m/s
- Impact time 0.3 sec (F=16288 N)

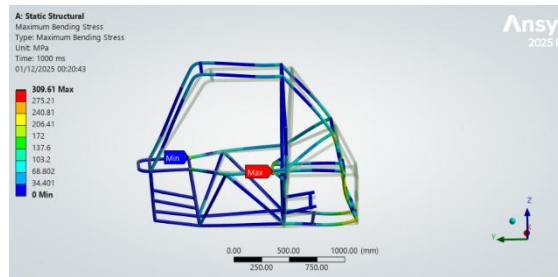
Boundary Condition

- The suspension points are fixed
- The force is applied on the rear most members of the chassis.



Boundary Condition

Total Deformation



Total Combined Stress

The Max Combined stress induced was 96.439 MPa. maximum deformation of 5.211 mm.

3. Side Impact

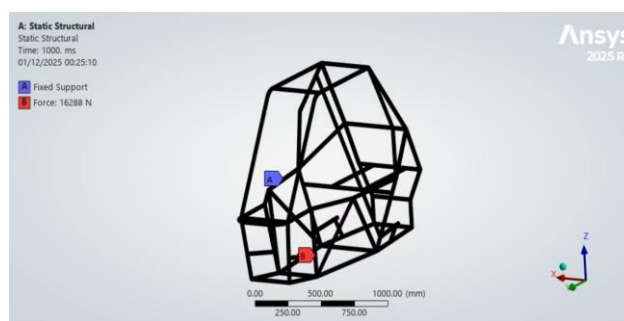
This analysis is done to simulate a scenario where the vehicle is hit by another vehicle at the side. Under such conditions, the forces act on the side members of the chassis.

Input Parameters:

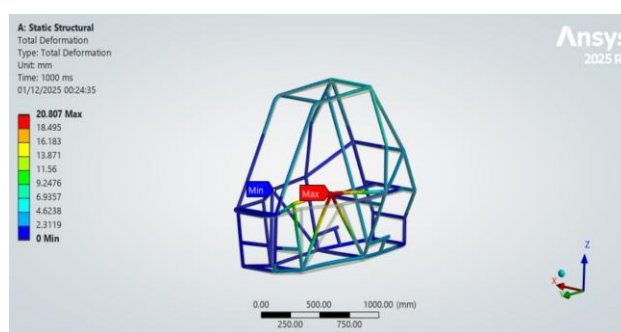
- weight=320 kg
- Speed =15.27m/s or 55km/hr.
- Impact time 0.3 s
- F=16288N

Boundary Condition:

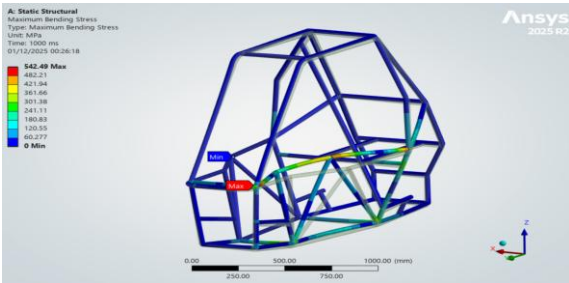
- The suspension points are fixed
- The force is applied on the side most members of the chassis



Boundary Conditions



Total Deformation



Total Combined Stress

The Maximum Combines stress induced is 157.28 MPa. maximum deformation of 2.7258 mm.

4. Rollover Impact

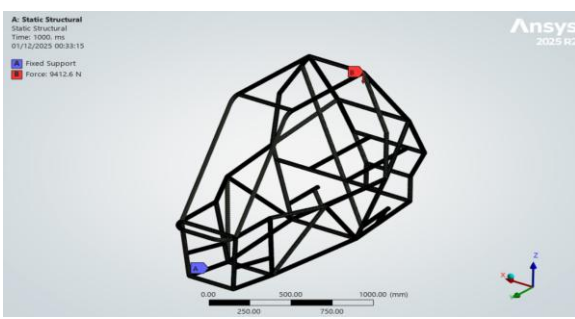
The vehicle is considered to be dropped from a height. Under such conditions, the forces act on the front most members of the chassis.

Input Parameters:

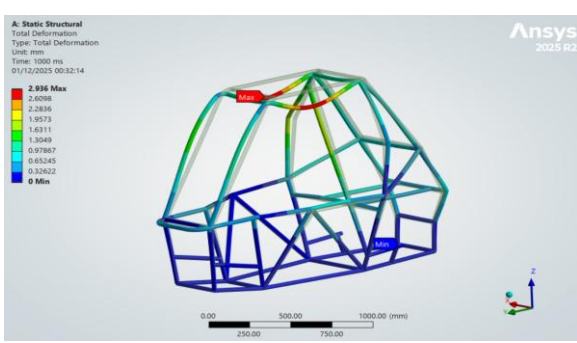
- Weight =320kg
- height= 10feet
- time= 0.15 s (F=9412.624)

Boundary Conditions:

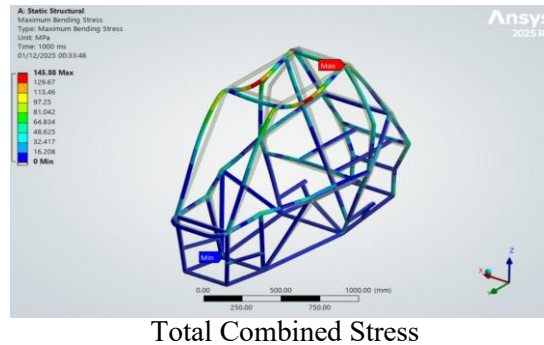
- The suspension points are fixed
- The force is applied on the top most members of the chassis



Boundary Condition



Total Deformation



Total Combined Stress

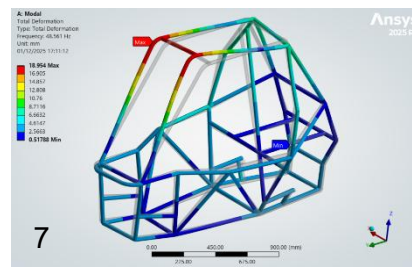
The Maximum stress induced by Maximum Combines is 68.154 MPa. deformation of 0.72276 mm.

Modal Analysis

This analysis was conducted on the roll cage to identify its natural frequencies and mode shapes to prevent catastrophic failure due to resonance.

The natural frequency of roll cage should not match with frequency of any vibrating component of vehicle in order to avoid the resonance.

Mode	Frequency (Hz)
1-6	0
7	48.561
8	51.966
9	63.22



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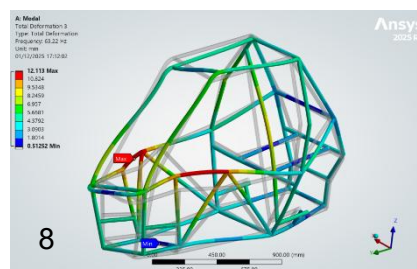

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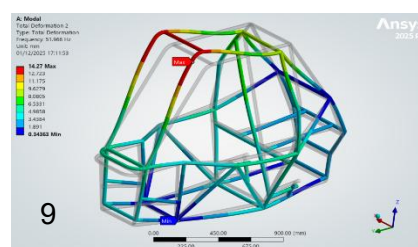

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Result & Conclusion

All this data can be used to conclude that the roll cage is fully capable of withstanding all types of impacts.

- **Front Impact:**
  Max stress = 448.96 MPa, deformation = 11.35 mm. the structure remains within acceptable limits for AISI 4130 tubing.

- **Rear Impact:**
  Max stress = 96.44 MPa, deformation = 5.21 mm. Stresses remained significantly below material limits, showing strong rear-end resistance.

- **Side Impact:**
  Max stress = 157.28 MPa, deformation = 2.73 mm. Side members demonstrated good stiffness with limited lateral deformation.

- **Rollover Impact:**
  Max stress = 68.15 MPa, deformation = 0.72 mm. Minimal deformation indicates high structural integrity during overturn conditions.

- **Modal Analysis:**
  The first natural frequency appears above 48 Hz, ensuring that the roll cage will not resonate with typical engine or terrain-induced vibrations.

Final Conclusion

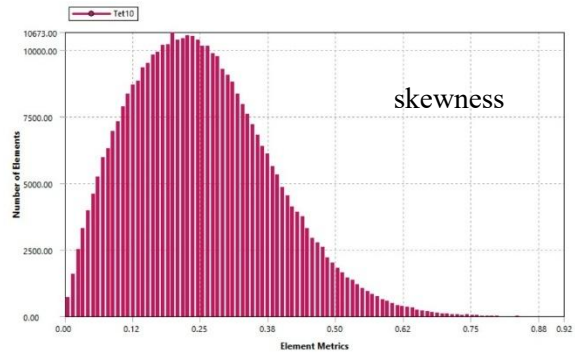
Overall, the roll cage design exhibits excellent structural strength, controlled deformation, and safe stress levels under all evaluated impact scenarios. The modal characteristics further ensure vibration safety.
Therefore, the roll cage is structurally robust and suitable for deployment in BAJA-style off-road conditions.

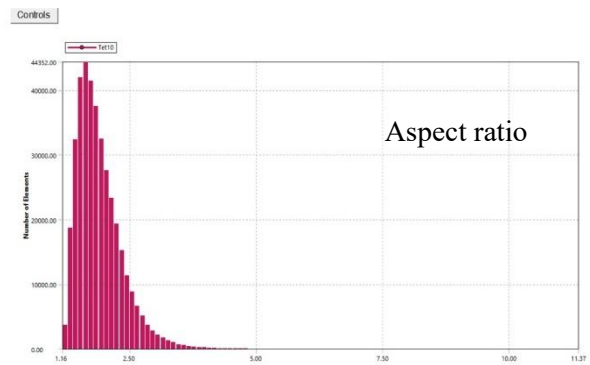
Thermal Analysis of Disc Brake

We performed this analysis on disc brake to evaluate its serviceability under high braking conditions

Mesh of Disc Brake

Proof of Element Quality





Metric	Result
Skewness	Majority between **0.15–0.35** (well below 0.9 limit) → **Good**
Aspect Ratio	Most elements between **1.1–2.0** (ideal range 1–3) → **Good**

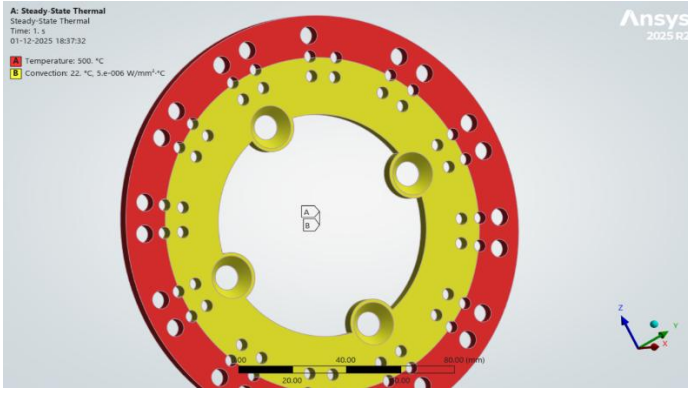
Both skewness and aspect ratio distributions indicate a very high-quality mesh suitable for accurate structural analysis.

Input Parameters

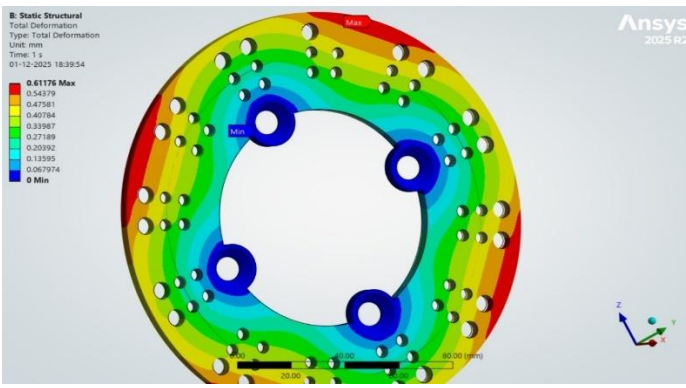
- Temperature applied on braking surface: 500 °C
- Ambient air temperature: 22 °C
- Convection coefficient: 6 W/m²·°C

Boundary Condition

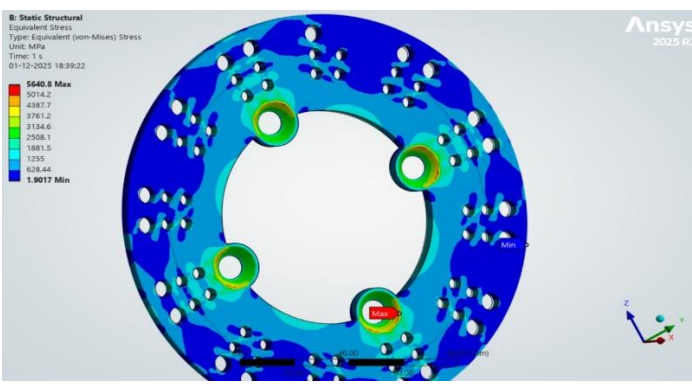
- Outer braking region to a 500 °C of fictional frictional braking.
- All remaining surfaces are subjected to natural convection at 22 °C to simulate airflow cooling.



Boundary condition



Total Deformation



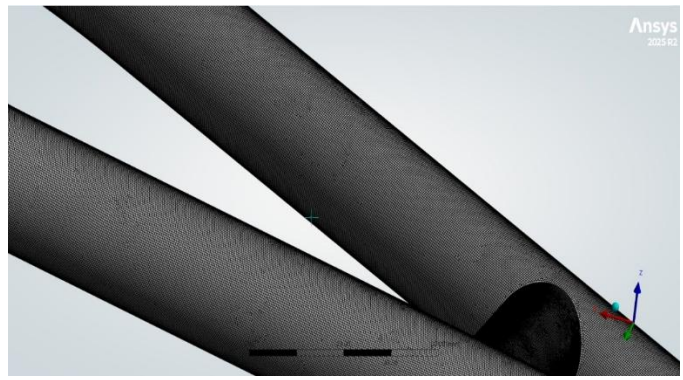
Total Stress

RESULT

The disc shows a maximum deformation of 0.61 mm with stress concentrated near the mounting holes, remaining within acceptable limits. Thermal loading produced a peak of 500 °C in simulation, representing a worst-case scenario to evaluate heat distribution and ensure structural safety under extreme braking conditions.

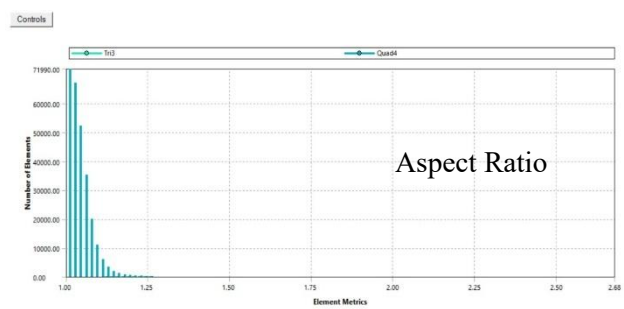
Control Arms Static Analysis

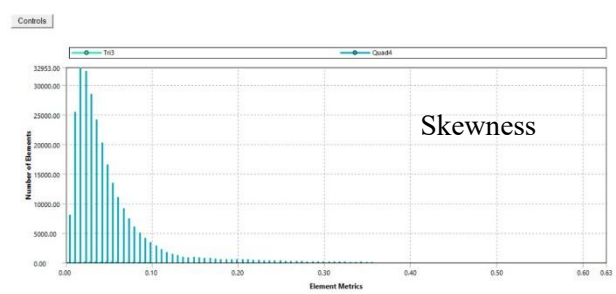
We performed static analysis on the control arms to evaluate their strength and stiffness under maximum suspension loads such as cornering, braking, and bumps. This ensures the arms can safely withstand peak forces without excessive deformation or structural failure.



Meshing

Proof of Element Quality





Metric	Result
Skewness	Majority between **0.02–0.12** (well below 0.9 limit) → **Good**
Aspect Ratio	Most elements between **1.0–1.15** (ideal range 1–3) → **Good**

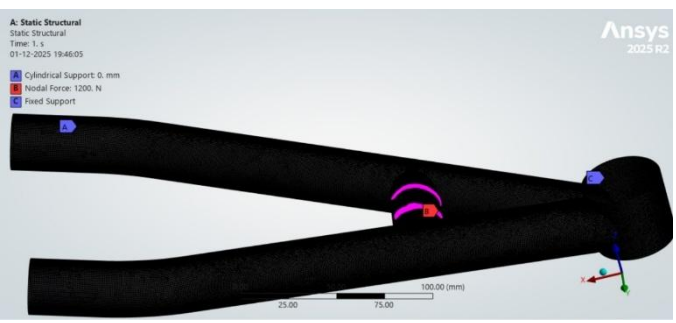
Both skewness and aspect ratio distributions indicate a very high-quality mesh suitable for accurate structural analysis.

Input Parameters:

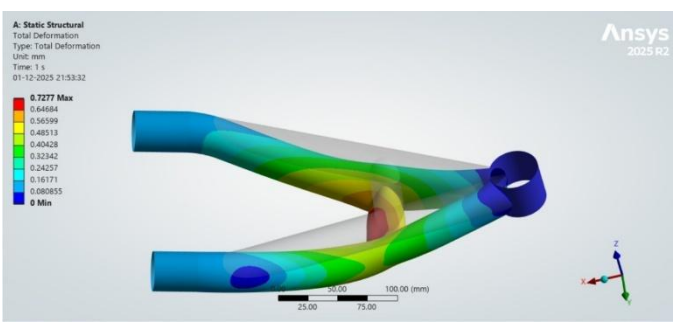
- Load Applied: 12,000 N
- Mass of the vehicle= 320 kg
- F=4G, therefore; 4x320x10 N= 12000N (approx.)

Boundary Condition:

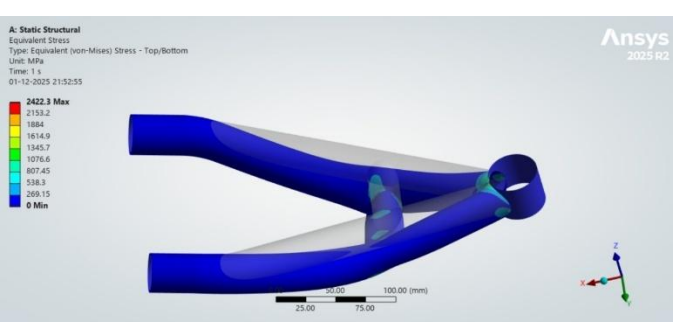
- A: Cylindrical support applied on the left tube
- C: Fixed support applied at the right end tube
- B: **12,000 N** nodal force applied on the inner cylindrical face at the joint



Meshing of Control Arm



Total Deformation

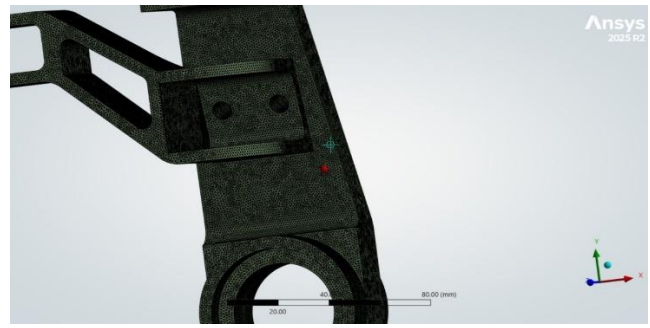


Total Stress induced

The component shows total deformation of only 0.072 mm under loading.
The peak von-Mises stress of 2422 MPa remains below the material yield strength, confirming structurally safe performance.

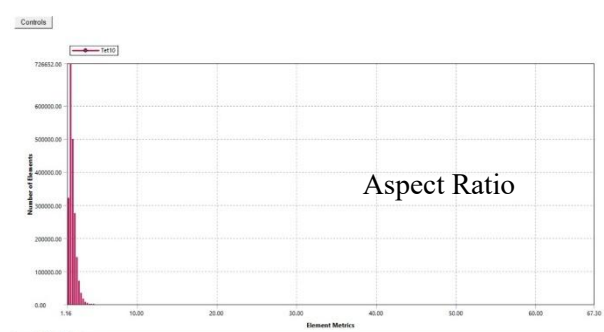
Upright Assembly Steering Analysis

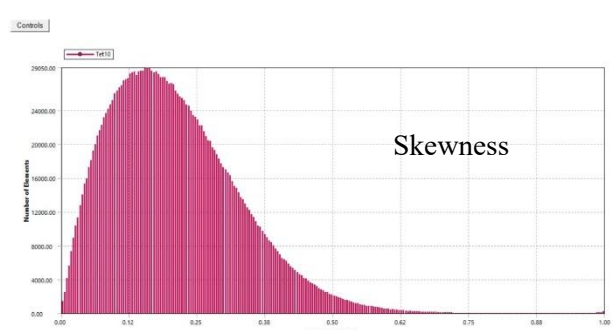
This analysis is done to evaluate how the upright behaves when steering forces are applied through the tie-rod. Under such conditions, the steering loads act directly on the upright, affecting its deformation, stress distribution, and overall steering response.



Meshing of upright

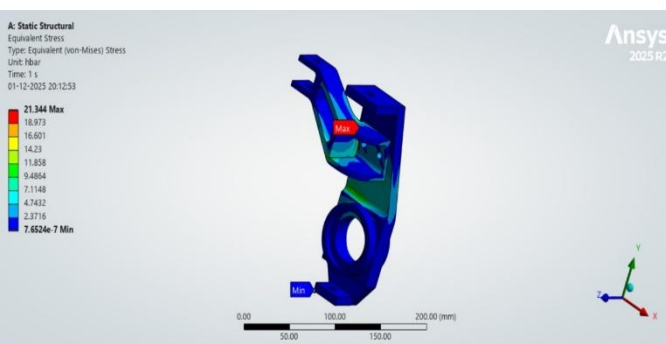
Proof of Mesh Quality



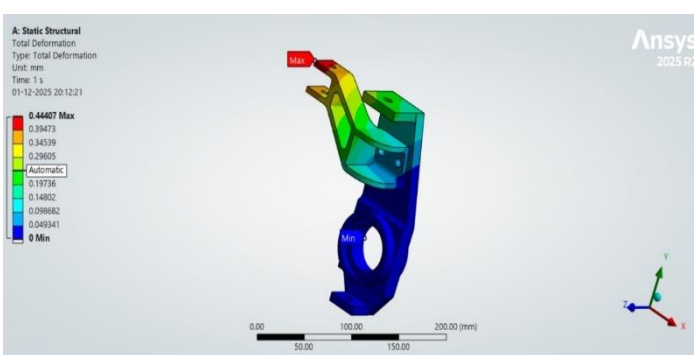


Metric	Result
Element Quality	Majority of elements clustered near **1.0–1.5** (well above minimum acceptable 0.1) → **Good**
Element Metric (2nd Plot)	Most elements between **0.10–0.35** (smooth distribution, no extremes) → **Good**

Boundary Condition



Total Stress

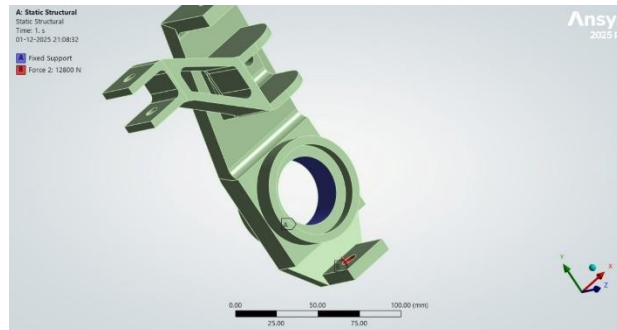


Total Deformation

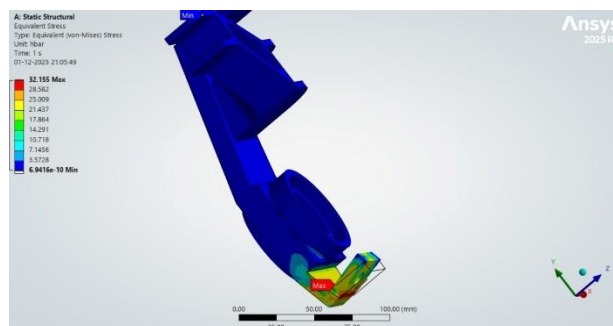
The upright experienced a maximum induced stress of 21.84 MPa and a maximum total deformation of 0.644 mm under the applied bumping load.

Upright Bump Analysis

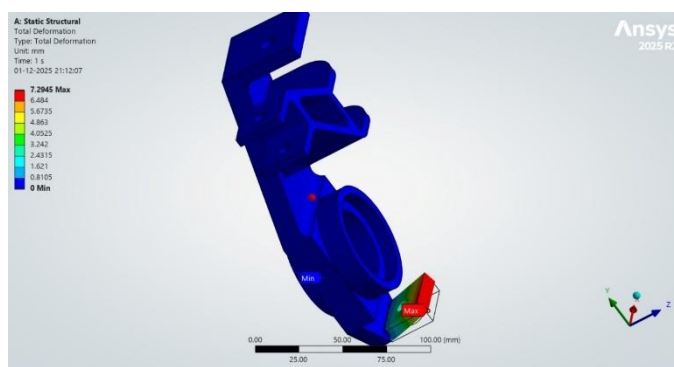
We performed this analysis to evaluate how the upright would behaves evaluate how the component responds when the wheel encounters vertical shocks from road bumps or obstacles.



Boundary Condition



Total Stress

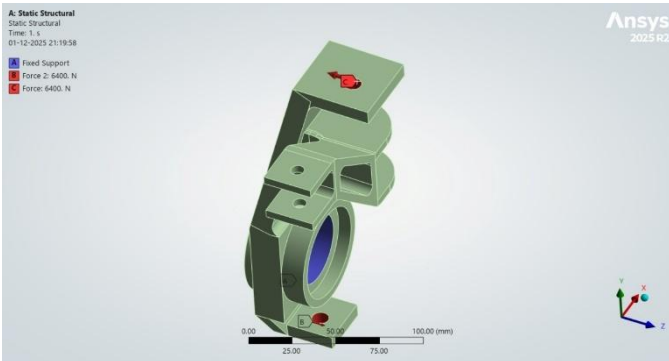


Total Deformation

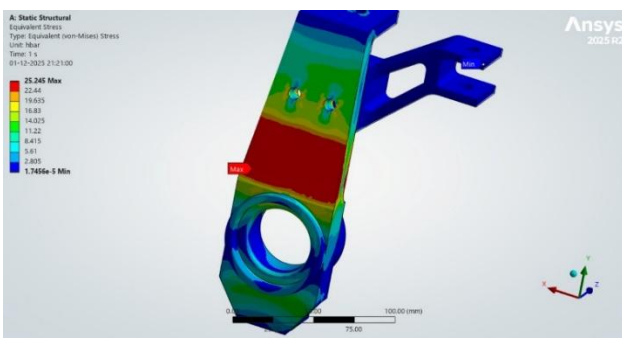
The upright experienced a maximum induced stress of 125.85 MPa and a maximum total deformation of 7.2845 mm under the applied load.

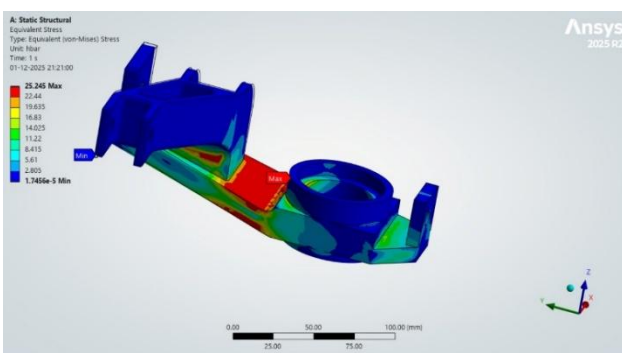
Upright cornering

We performed this analysis to evaluate how the upright would behaves evaluate how the component responds when the wheel encounters turning.

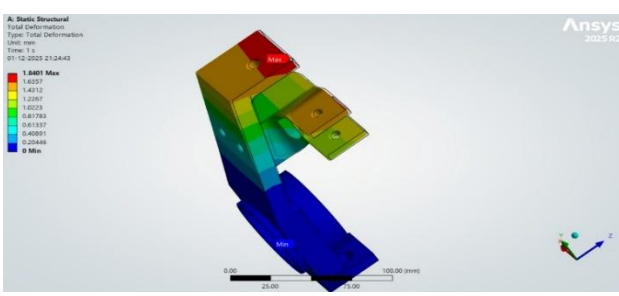


Boundary Condition





Total Stress



Total Deformation

Result

Case	Max Stress (MPa)	Max Deformation (mm)	Mesh Quality	Remarks
Steering	21.84	0.644	Good	Safe
Bump	125.85	7.2845	Good	Highest load case
Cornering	Not provided	Not provided	Good	Stress at expected zones

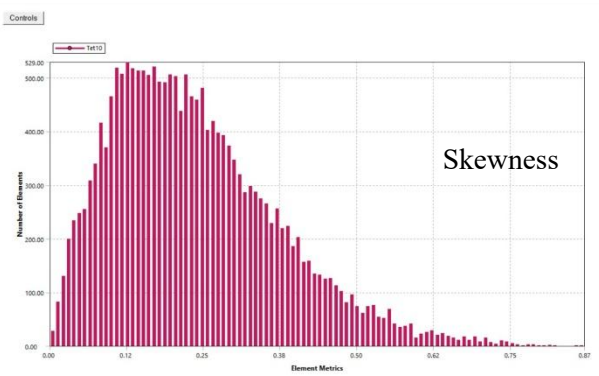
The upright remains structurally safe under steering loads, exhibiting low stress and negligible deformation localized at the steering arm. Bump loading produces the highest stress and displacement, primarily around the calliper mount and upright body, making it the governing case for design refinement, though still within acceptable limits. Cornering loads show expected stress concentration near the steering arm–hub interface, but incomplete numerical data limits full validation. Overall, the upright demonstrates acceptable performance across all evaluated load cases, with bump loading identified as the critical condition.

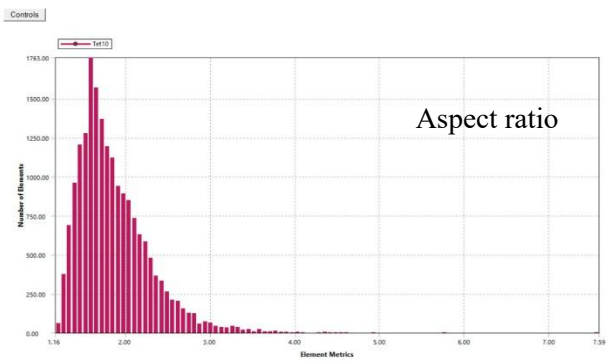
Static Analysis of Brake Pedal

We performed Static Analysis on brake pedal in compliance with the rulebook and assess how it would perform under sudden impact/force and would not brake bend or deform.

Proof of Mesh Quality

Here is the graph of skewness and aspect ratio charts of out brake pedal giving insight on our geometry.





Metric	Result
Skewness	Most elements in 0.10–0.35 (well below 0.5) → Good
Aspect Ratio	Majority in 1.0–2.5; very few >4 → Good

Boundary Conditions

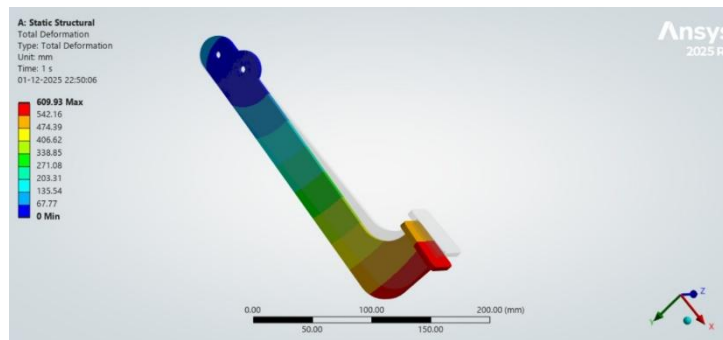
The Brake pedal was fixed on point B and a force of 2000 N was applied on the foot rest.

Input Parameters

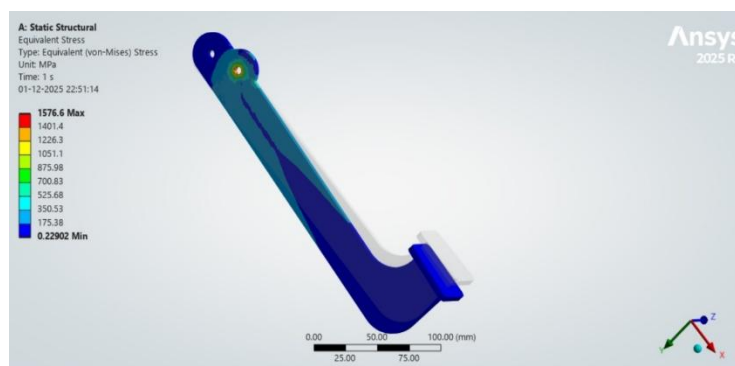
The input parameter of the brake pedal was in compliance with rulebook.

The material of the brake pedal is AISI 1010 low carbon steel.

Boundary Condition



Total Deformation



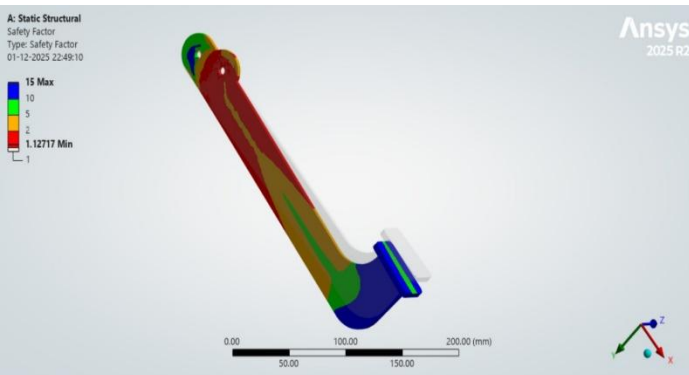
Total Stress

The analysis shows a maximum stress of about 1576 MPa concentrated at the upper joint, indicating that the component is overstressed and likely to fail at that location. The maximum deformation reaches approximately 610 mm at the free end, which is excessively high and structurally unacceptable. Overall, the part is unable to sustain the applied load in its current design and requires geometric reinforcement or correction of loading/boundary conditions.

Hence, we plan to switch the material of the brake padel or increase its cross section.

Safety Factor of Brake Pedal

The safety factor distribution shows a minimum value of about 1.12 near the loaded joint, indicating the component is operating very close to its allowable limit, while the rest of the structure remains in a safer range. Overall, the design provides marginal safety and would benefit from reinforcement in the high-stress region.



Result & Conclusion

The structural integrity of the ATV was assessed using SolidWorks 2025 for geometric modelling and ANSYS Workbench 2025 R2 for finite-element analysis. The primary objective of the study was to validate occupant safety and ensure the durability of critical components under competitive loading scenarios.
All major subsystems; such as the roll cage, uprights, and control arms. They demonstrated compliance with the prescribed safety factors and exhibited satisfactory stress and deformation responses. In contrast, the brake pedal assembly showed a localized structural failure, indicating insufficient load-bearing capability and necessitating immediate design modification and reinforcement.

Summary of each system

A. Roll Cage (Chassis)

The roll cage, fabricated from AISI 4130 Chromoly tubing, exhibited robust structural performance across all simulated loading and impact conditions.

Impact Resistance: All direct and multi-directional impact scenarios remained within allowable stress limits. Rear-impact loading demonstrated exceptionally high structural resilience, with stresses well below the material yield threshold.

Vibrational Safety: Modal analysis identified the first natural frequency at 48.561 Hz, sufficiently above engine firing and terrain-induced excitation frequencies. This effectively eliminates the risk of resonant amplification and fatigue-driven structural failure.

B. Suspension & Steering Subsystems

Control Arms: Under a peak applied load of 12,000 N (representing a 4G event), the control arms demonstrated high stiffness, with a maximum deformation of 0.072 mm, confirming adequate strength against dynamic suspension loads.

Uprights: Evaluation under steering, cornering, and bump loads identified bump loading as the governing case. The component experienced a peak stress of 125.85 MPa and deformation of 7.28 mm, both within acceptable design margins, ensuring operational safety.

C. Braking System

Disc Brake: Thermal-structural coupling at a worst-case rotor temperature of 500 °C showed localized stress concentrations around the mounting interface. However, the maximum deformation was limited to 0.61 mm, indicating stable thermal performance.

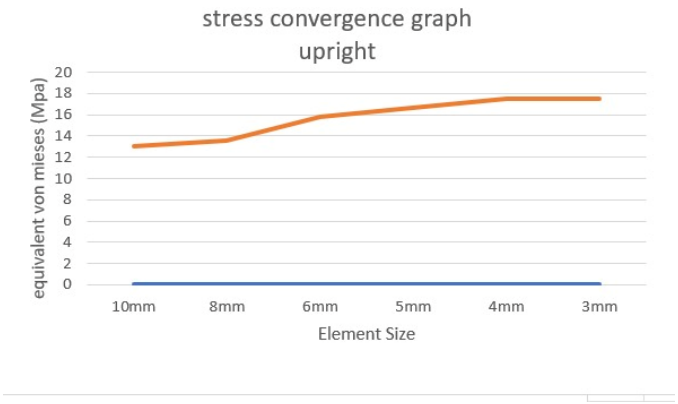
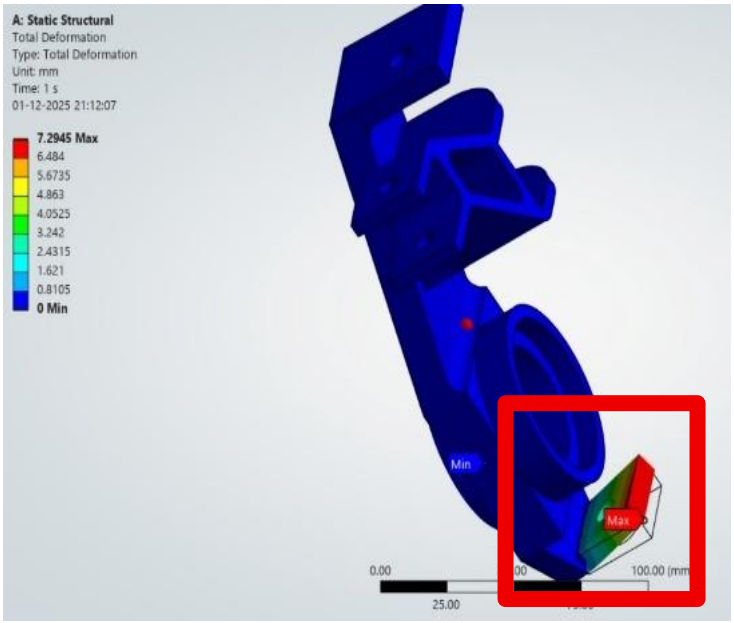
Brake Pedal (Critical Weakness): Static loading of the AISI 1010 steel pedal under 2000 N revealed a critical structural deficiency.

Maximum Stress: 1576 MPa, concentrated at the upper hinge connection.

Deformation: Approximately 610 mm at the free end far exceeding acceptable displacement thresholds.

Safety Factor: The minimum safety factor of 1.12 indicates the pedal is operating dangerously close to failure, necessitating immediate redesign and reinforcement.

Our Analysis of Upright Bump came to real life during our physical testing of the ATV shown below.

Stress Convergence Summary – Upright Component

The Von Mises stress values increase and gradually stabilize as the mesh is refined, with negligible change beyond a 4 mm element size. This indicates that the solution has converged, and the 4–3 mm mesh range provides an accurate and reliable stress prediction for the upright.